

8 Transport in mammals

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Double circulation sends blood through pulmonary and systemic circuits; separation permits high systemic pressure and lower pulmonary pressure. Arteries have thick elastic walls, veins valves and larger lumens; capillaries have thin walls for exchange. Haemoglobin's cooperative oxygen binding produces a sigmoidal dissociation curve; a lower affinity in metabolically active tissue assists unloading.

Worked example: follow the reasoning

A tissue with higher CO_2 shifts haemoglobin's dissociation curve to the right (Bohr effect), promoting oxygen release at a given oxygen partial pressure.

Quick check · multiple choice

A What is the Bohr effect useful for in active tissue? [1]

- A More blood clotting
- B Lower CO_2
- C No diffusion
- D Oxygen unloading

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Explain two structural adaptations of a capillary for exchange and why oxygen unloading is greater in active tissues. [3]

Answers and explanation

Quick check: D. Oxygen unloading

Question 1 [3 marks]

Thin endothelial wall minimises diffusion distance [1]; narrow lumen/slower flow or extensive network increases exchange time/surface [1]; high CO₂ lowers haemoglobin affinity (Bohr effect) [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Biology 9700, 2025–2027. <https://www.cambridgeinternational.org/Images/664560-2025-2027-syllabus.pdf>

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